

Echomax

A multi-mode delay line: tape, bucket-brigade, clean digital and a driven kit-built echo in one box.



Overview

Echomax is four echoes in one. Each type has its own sound in the feedback loop, so the repeats change character as they circulate: TAPE saturates warmly and darkens, BUCKET is very dark and gritty with a slow clock wobble, DIGITAL stays clean, and ECHORDER drives an op-amp hard into a dark bucket-brigade filter with a little background hiss.

The delay time can be set in milliseconds or locked to the project tempo. When you change it, the repeats glide to the new time rather than jumping - slowly on TAPE (for the classic pitch swoop), quickly on DIGITAL.

Around the echo sit ping-pong, freeze, ducking, a vintage 500 Hz cut on the echoes, drift for wow and flutter, and the IN / OUT strip with trim, saturation and a clipper or limiter.

Reach for it for

- Slapback and tape echo on vocals and guitars
- Dub echoes that build into self-oscillation
- Clean, ducked vocal delays that stay out of the way
- Ping-pong echoes across the stereo field
- Freezing a phrase into a loop

Controls

Echo types (header)

TAPE	Warm tape saturation and a high-frequency roll-off (about 5.5 kHz) on every pass. Time changes glide slowly, so turning TIME gives the tape pitch swoop.
BUCKET	A bucket-brigade echo: a steep, dark roll-off (about 2.8 kHz), gritty clipping in the loop and a slow clock wobble on the delay.
DIGITAL	Clean and uncoloured: DRIVE has no effect (the knob reads clean), and only very hot repeats are softly limited.
ECHORDER	A driven kit-built echo: an overdriven op-amp with lopsided saturation, a dark bucket-brigade filter (about 3.2 kHz), a slow clock drift and background hiss that rises with DRIVE. DUAL CHIP works only in this mode.

Readout

Time display	The delay time in ms (also when synced), with the sync division or FREE beside it, and FROZEN, RUNAWAY (feedback above 100%) or PING-PONG when they apply.
Repeat ladder	Sixteen lights showing how the repeats die away at the current feedback. With ping-pong on, the repeats alternate between two rows.

Switches

SYNC	Locks TIME to the project tempo (on by default).
PING	Ping-pong: the input is summed to mono and the repeats bounce between left and right.
FREEZE	Stops recording new input and keeps what is in the delay circulating without loss - an endless loop you can play over.

DUAL CHIP	ECHORDER only: the right side runs off its own slightly mistuned clock, so with WIDTH up it drifts and beats against the left instead of sitting at a fixed offset.
VINTAGE EQ (OFF / -3dB / -6dB / -9dB)	The classic studio trick on the echoes only: a cut at 500 Hz of the chosen depth, with a high-pass at 140 Hz and a low-pass at 5 kHz. OFF bypasses it.

Delay

Hover over a knob to see its value.

TIME	The delay time, 20 to 2000 ms, or with SYNC a note value from 4 bars to 1/32 (dotted and triplet included).
FEEDBACK	How much of each repeat is fed back, 0 to 110%. Past 100% the echo grows on its own (RUNAWAY) - the loop's saturation keeps it from exploding.
DRIVE	How hard the repeats are pushed into the type's saturation, 0 to 100%. Not used in DIGITAL.
LO CUT	A high-pass inside the feedback loop, 20 to 1000 Hz: each repeat gets thinner than the last.
HI CUT	A low-pass inside the feedback loop, 1 to 20 kHz: each repeat gets darker than the last.
DUCK	Turns the echoes down while you play, 0 to 100%, and lets them come back up in the gaps (fast to dip, about a quarter of a second to recover).

Drift

Tape-style wow (a slow random wander of the delay time) and flutter (a fast wobble around 6.5 Hz). The amounts of wow and flutter come with the presets.

DRIFT	0 to 100%. 50% gives the standard amount of wow and flutter, 100% doubles it, 0 turns it off. The wander also speeds up as you turn it up.
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IN / OUT

TRIM	Input level before anything else, from -24 to +12 dB.
GUARD	Off by default. Click the TRIM knob (without dragging) to switch it; the small LED at the knob's upper right shows it is on and glows brighter while it works. It gently eases very hot sources down towards about -18 dBFS average and catches fast peaks above -6 dBFS. It never boosts quiet sources.
SAT	Band saturation on the way out, 0 to 100%: only the lows and highs are saturated (the mids stay clean), with a soft curve that loosens as the level rises.
OFF / CLIP / LIMIT (OUT)	The final stage: nothing, a soft clipper (no latency) that holds the peaks at -0.3 dBFS, or a lookahead limiter (3 ms latency, compensated by REAPER) that holds them at -0.3 dBFS. The CLIP key flashes red while the clipper is working.

Master

MIX	Dry / echo balance, 0 to 100% (35% by default). At 50% both are at full level; below that the echoes fade, above it the dry signal fades.
WIDTH	The stereo spread of the echoes, 0 to 100%. At 0% the echoes are mono; turning it up keeps the difference between the sides and makes the right side's delay a touch longer than the left.
OUTPUT	Output level, -24 to +12 dB.
OUT meters	Output level, left and right. Click to reset the peak hold.

The header

PRESETS	Opens the list of factory presets; the arrows beside it step through them, and the display shows the one in use (click it
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	for the list too). The last entry resets everything to the defaults.
S / M / L	Panel size: small, medium or large. The panel also scales to fit the plugin window.
?	Opens a pop-up with the design credit, the GFX website address and the address of this manual. Click anywhere to close it.

Presets

The presets set the echo type, time, feedback, drive, filters, duck, ping-pong, wow, flutter, mix and drift, and switch FREEZE off; the Echorder dub presets also set DUAL CHIP. WIDTH, OUTPUT, VINTAGE EQ and the IN / OUT strip are left alone.

Tape: Worn Tape Echo	Dotted-eighth tape echo with heavy wow and plenty of drift.
Tape: Slapback Saturation	A 110 ms tape slap with saturation.
Tape: Dub Siren Oscillation	Quarter-note tape echo at 85% feedback, dark and wobbly, on the edge of oscillation.
Bucket: Memory Lane Ping-Pong	Eighth-note bucket-brigade echoes bouncing left and right, lightly ducked.
Bucket: Dark Analog Crunch	180 ms dark, driven bucket echoes with strong feedback.
Echorder: EC-444 Drive	280 ms driven Echorder repeats at 70% feedback.
Echorder: Slap & Crunch	An 85 ms slap pushed to near-full drive.
Digital: Clean Ducked Vocal	Clean quarter-note echo, ducked by 50%, no wow or flutter.
Digital: Crystal Stereo Triplet	Clean eighth-note-triplet ping-pong, open to 18 kHz.

Tape: Ambient Flutter Wash	A 1.2 s tape echo with 75% feedback, ducking and lots of wow - an ambient wash.
Echorder: Self-Oscillating Synth	Dotted-eighth Echorder at 102% feedback: it keeps building on its own.
Echorder: Deep Dub Echo	380 ms dark dub echo, single chip.
Echorder: Modern Dub Wash	300 ms with more wow, DUAL CHIP on for a drifting stereo image.
Echorder: Chaos Dub Grit	220 ms, 85% feedback, very dark and gritty, single chip.
Echorder: Heavyweight Dub Bass	450 ms dark dub echo with wow, DUAL CHIP on.
Reset Defaults	Every control back to its starting value.

Tips

- On TAPE, sweep TIME during a repeat for the classic pitch-bending dub swoop.
- For a vocal echo that never clutters the words, try DIGITAL with DUCK at 40 - 60%.
- Push FEEDBACK past 100% and ride HI CUT or FREEZE for dub build-ups; pull it back before it gets too loud.
- VINTAGE EQ at -6 dB takes the boxiness out of the repeats so they sit behind the dry sound.

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